# IMO 2026 — Problem 6

Solution by GPT-5.6 Sol (xhigh reasoning)

2026-07-18

**Source**: `results/gpt-5.6-sol-xhigh/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

## Problem

Let $a_1, a_2, a_3, \ldots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers $n$, the number $a_{n+1}$ is the smallest positive integer greater than $a_n$ such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \ldots, n$. Prove that there exist positive integers $T$ and $L$ such that $a_{n+T} = a_n + L$ for every positive integer $n$. (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers $x$ and $y$.)

## Solution document

### Status

solved

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### Approaches tried

- Reformulated the sequence as the result of scanning all integers above $a_1$ and retaining exactly those non-coprime to every previously retained integer.
- A direct attempt to bound all prime divisors of the terms fails: terms can have arbitrarily large irrelevant prime factors.
- The successful route is instead to consider the inclusion-minimal prime supports of retained integers. They form a self-blocking intersecting family. An arithmetic descent shows that every prime occurring in a minimal support also occurs in a minimal support whose remaining radical is at most $a_1$. There are only finitely many such small cores, and a disjoint witness

shows that each core can be paired with only finitely many primes. Hence the set of relevant primes is finite.

## Current best

Let $\mathcal{H}$ be the family of inclusion-minimal prime supports of sequence terms. The proof establishes that $U = \bigcup \mathcal{H}$ is finite. Consequently the sequence is exactly the increasing enumeration, beginning at $a_1$, of integers divisible by $\prod_{p \in H} p$ for at least one $H \in \mathcal{H}$. This condition is periodic modulo $L = \prod_{p \in U} p$. If $T$ is the number of admissible residue classes modulo $L$, then $a_{n+T} = a_n + L$ for every $n$.

## Full proof

Put $m = a_1$, and for every integer $x > 1$ let $P(x)$ denote the finite set of prime divisors of $x$.

### 1. The self-blocking family of prime supports

The terms $a_n$ are strictly increasing, so the sequence is unbounded. We first record the following consequence of the defining rule:

> **Scanning property.** If an integer $x > m$ is not a term of the sequence, then there is a term $a_i < x$ such that $\gcd(x, a_i) = 1$.

Indeed, choose $n$ such that

$$a_n < x < a_{n+1}.$$

If $x$ had a common prime divisor with each of $a_1, \ldots, a_n$, it would be an eligible choice after $a_n$, contradicting the fact that $a_{n+1}$ is the smallest eligible integer. This proves the scanning property.

Let

$$\mathcal{G} = \{P(a_n) : n \geq 1\}$$

and define its *blocker* (among finite sets of primes) by

$$\mathcal{F} = \{S : S \text{ is a finite set of primes and } S \cap G \neq \emptyset \text{ for every } G \in \mathcal{G}\}.$$

Any two terms of the sequence have gcd greater than 1, so any two members of $\mathcal{G}$ intersect. It follows that $\mathcal{G} \subseteq \mathcal{F}$.

In fact,

$$\boxed{\mathcal{G} = \mathcal{F}.} \qquad (1)$$

To prove the reverse inclusion, take $S \in \mathcal{F}$. The set $S$ is nonempty, because it must intersect $P(a_1)$. Choose an integer $x > m$ with $P(x) = S$; for example, a sufficiently large power of $\prod_{p \in S} p$ has these properties. Since $S$ intersects every member of $\mathcal{G}$, the integer $x$ has gcd greater than 1 with every term of the sequence. As the sequence is unbounded, we may choose $n$ such that

$$a_n < x \leq a_{n+1}.$$

Then $x$ is eligible after $a_n$, so the minimality in the recurrence gives $a_{n+1} \leq x$. Hence $x = a_{n+1}$, and therefore $S = P(x) \in \mathcal{G}$. This proves (1).

Thus $\mathcal{F}$ has the following properties:

1. it is **upward-closed**: if $S \in \mathcal{F}$ and $S \subseteq S'$, where $S'$ is finite, then $S' \in \mathcal{F}$;
2. it is **intersecting**: any two of its members intersect;
3. it is **self-blocking**:

$$S \in \mathcal{F} \quad \iff \quad S \cap F \neq \emptyset \text{ for every } F \in \mathcal{F}. \qquad (2)$$

Here (2) follows from the definition of $\mathcal{F}$ and (1). In particular, whenever a finite set $S$ does not belong to $\mathcal{F}$, there exists $F \in \mathcal{F}$ disjoint from $S$.

Let $\mathcal{H}$ be the family of inclusion-minimal members of $\mathcal{F}$. Every member $F$ of $\mathcal{F}$ contains a member of $\mathcal{H}$: among the finitely many subsets of $F$ which belong to $\mathcal{F}$, choose an inclusion-minimal one.

## 2. Finiteness of the relevant primes

Call the primes in

$$U = \bigcup_{H \in \mathcal{H}} H$$

the **relevant primes**. We prove that $U$ is finite.

Fix $p \in U$, and choose $H \in \mathcal{H}$ containing $p$. Set

$$c(H,p) = \prod_{q \in H \setminus \{p\}} q,$$

where an empty product is 1. We first prove that, after possibly replacing $H$ by another member of $\mathcal{H}$ which still contains $p$, we may arrange that

$$c(H,p) \leq m. \qquad (3)$$

Suppose that the present $H$ does not satisfy (3), and put

$$D = H \setminus \{p\}, \qquad x = c(H,p) > m.$$

The minimality of $H$ gives $D \notin \mathcal{F}$. By (1), no term of the sequence has prime support $D$; in particular, $x$ is not a term. By the scanning property there is a term $y = a_j < x$ with

$$P(y) \cap D = \emptyset.$$

The set $P(y)$ belongs to $\mathcal{F}$, so it contains some $K \in \mathcal{H}$. Hence $K \cap D = \emptyset$. On the other hand, $K, H \in \mathcal{F}$, and $\mathcal{F}$ is intersecting. Since $H = D \cup \{p\}$, these two facts force $p \in K$. Moreover, the squarefree number $p\, c(K,p)$ divides $y$, and therefore

$$c(K,p) \leq \frac{y}{p} < x = c(H,p). \qquad (4)$$

Thus replacing $H$ by $K$ strictly decreases the positive integer $c(H,p)$ while preserving both $H \in \mathcal{H}$ and $p \in H$. Repeating this operation must terminate, and it can terminate only with (3). This proves the claim.

There are only finitely many finite sets $C$ of primes satisfying

$$\prod_{q \in C} q \leq m, \qquad (5)$$

because every prime in such a set is at most $m$. By (3), every $p \in U$ admits at least one such set $C$ for which

$$C \cup \{p\} \in \mathcal{H}, \qquad p \notin C. \qquad (6)$$

We show that each fixed $C$ satisfying (5) can occur in (6) for only finitely many primes $p$.

If (6) holds, the inclusion-minimality of $C \cup \{p\}$ implies that $C \notin \mathcal{F}$. By (2), we may therefore fix some $F_C \in \mathcal{F}$ with

$$F_C \cap C = \emptyset.$$

For every prime $p$ satisfying (6), the two members $F_C$ and $C \cup \{p\}$ of the intersecting family $\mathcal{F}$ must intersect. Since $F_C$ is disjoint from $C$, this forces $p \in F_C$. The set $F_C$ is finite, so only finitely many such $p$ exist.

There are finitely many possible $C$, and each yields only finitely many possible $p$. Hence $U$ is finite. Since every $H \in \mathcal{H}$ is a subset of the finite set $U$, the family $\mathcal{H}$ itself is finite as well.

### 3. Periodicity

For every finite set $S$ of primes, upward-closure and the definition of $\mathcal{H}$ give

$$S \in \mathcal{F} \iff H \subseteq S \text{ for some } H \in \mathcal{H}. \qquad (7)$$

We now characterize the terms of the sequence. For every integer $x \geq m$,

$$x \text{ is a term of the sequence} \iff H \subseteq P(x) \text{ for some } H \in \mathcal{H}. \qquad (8)$$

The forward implication follows from $P(x) \in \mathcal{G} = \mathcal{F}$ and (7). Conversely, suppose the right-hand side holds. Then $P(x) \in \mathcal{F}$, so $x$ has gcd greater than 1 with every term of the sequence. If $x = m$, it is already the first term. If $x > m$ and were not a term, the scanning property would supply an earlier term coprime to $x$, a contradiction. This proves (8).

The set $U$ is nonempty because $\mathcal{F}$ contains $P(m)$ and hence $\mathcal{H}$ is nonempty. Define

$$L = \prod_{p \in U} p.$$

For any positive integer $x$ and any $H \in \mathcal{H}$, all primes of $H$ divide $L$, so

$$H \subseteq P(x) \iff H \subseteq P(x + L).$$

Thus the condition on the right of (8) is periodic modulo $L$.

Let $T$ be the number of integers $r \in \{1, 2, \dots, L\}$ satisfying that condition. This is positive: the integer $L$ itself satisfies it because it is divisible by every prime in every $H \in \mathcal{H}$. By periodicity, every interval of $L$ consecutive integers contains exactly $T$ integers satisfying the condition.

Fix $n \geq 1$. By (8), among the integers

$$a_n + 1, a_n + 2, \dots, a_n + L$$

there are exactly $T$ terms of the sequence. Also $a_n + L$ is one of them, since $a_n$ is a term and the condition in (8) is periodic modulo $L$. Therefore $a_n + L$ is precisely the $T$-th term after $a_n$, that is,

$$\boxed{a_{n+T} = a_n + L}$$

for every positive integer $n$. Both $T$ and $L$ are positive, completing the proof.